

Lecture 5

CURVILINEAR MOTION: NORMAL AND TANGENTIAL COMPONENTS

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CURVILINEAR MOTION: NORMAL AND TANGENTIAL COMPONENTS

Today's Objectives:

Students will be able to:

1. Determine the normal and tangential components of velocity and acceleration of a particle traveling along a curved path.



In-Class Activities:

- Reading Quiz
- Applications
- Normal and Tangential Components of Velocity and Acceleration
- Special Cases of Motion
- Example Problems
- Concept Quiz
- Group Problem Solving
- Attention Quiz

READING QUIZ

1. If a particle moves along a curve with a constant speed, then its tangential component of acceleration is
A) positive. B) negative.
C) zero. D) constant.
2. The normal component of acceleration represents
A) the time rate of change in the magnitude of the velocity.
B) the time rate of change in the direction of the velocity.
C) magnitude of the velocity.
D) direction of the total acceleration.

APPLICATIONS

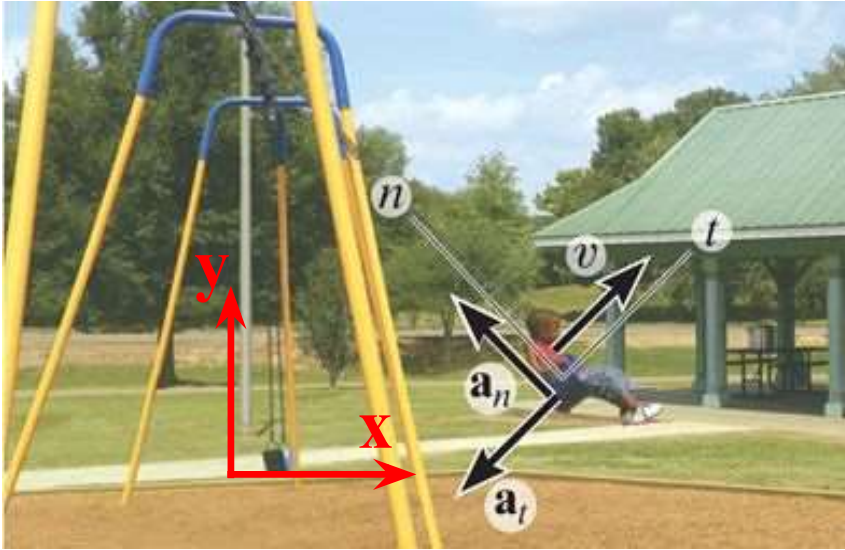


Cars traveling along a clover-leaf interchange experience an acceleration due to a change in velocity as well as due to a change in direction of the velocity.

If the car's speed is increasing at a known rate as it travels along a curve, how can we determine the magnitude and direction of its total acceleration?

Why would you care about the total acceleration of the car?

APPLICATIONS (continued)



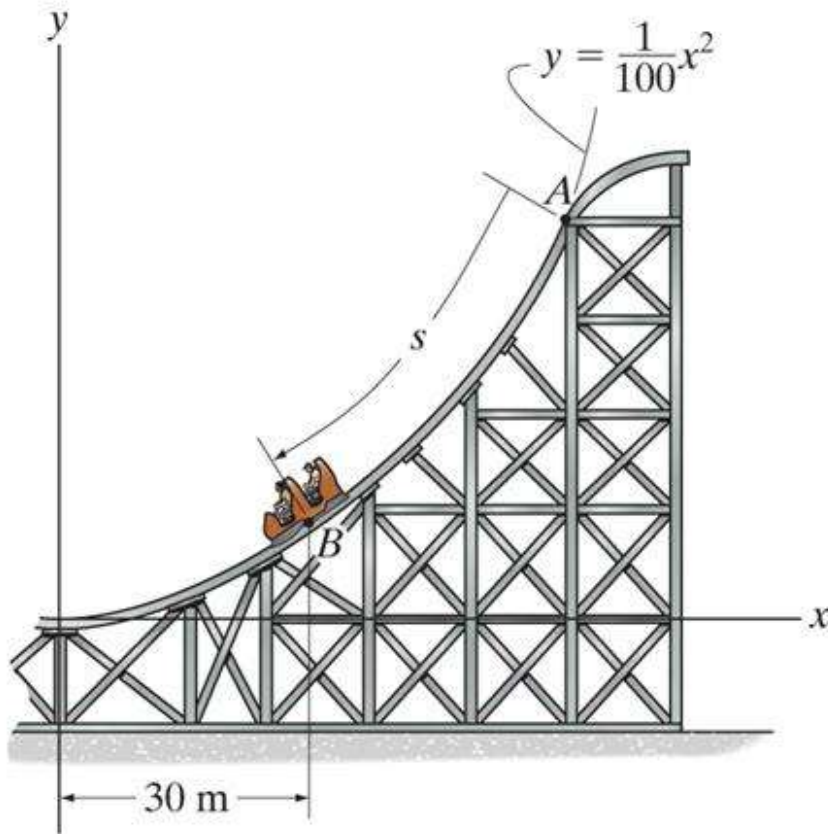
As the boy swings upward with a velocity $\mathbf{v}$, his motion can be analyzed using n - t coordinates.

As he rises, the magnitude of his velocity is changing, and thus his acceleration is also changing.

How can we determine his velocity and acceleration at the bottom of the arc?

Can we use different coordinates, such as x - y coordinates, to describe his motion? Which coordinate system would be easier to use to describe his motion? Why?

APPLICATIONS (continued)



A roller coaster travels down a hill for which the path can be approximated by a function $y = f(x)$.

The roller coaster starts from rest and increases its speed at a constant rate.

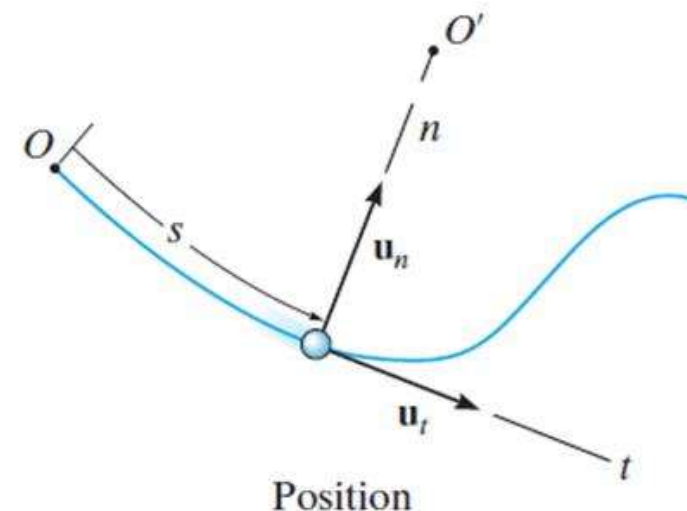
How can we determine its velocity and acceleration at the bottom?

Why would we want to know these values?

NORMAL AND TANGENTIAL COMPONENTS (Section 12.7)

When a particle moves along a curved path, it is sometimes convenient to describe its motion using coordinates other than Cartesian. When the **path of motion is known**, **normal (n)** and **tangential (t)** coordinates are often used.

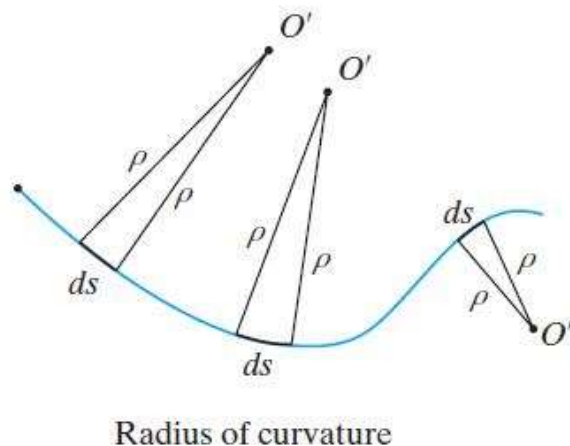
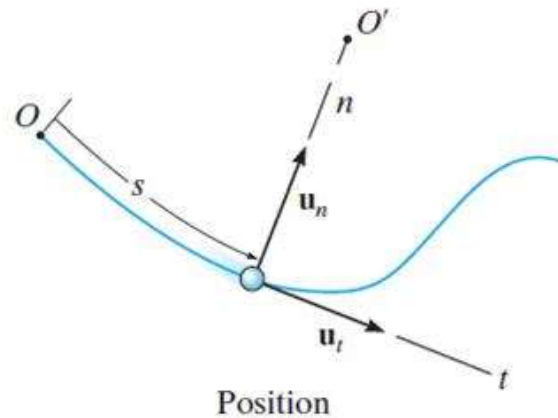
In the n-t coordinate system, the **origin is located on the particle** (thus the origin and coordinate system **move with the particle**).



The **t-axis** is **tangent** to the **path (curve)** at the instant considered, positive in the direction of the particle's motion.

The **n-axis** is **perpendicular** to the **t-axis** with the positive direction toward the center of curvature of the curve.

NORMAL AND TANGENTIAL COMPONENTS (continued)

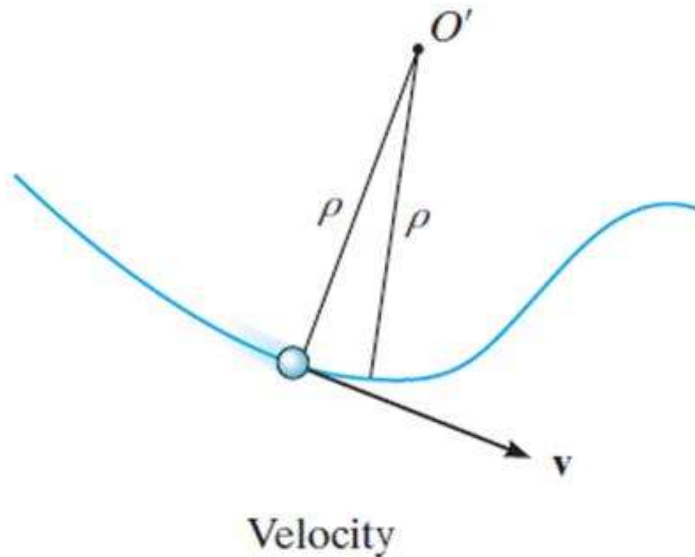


The positive n and t directions are defined by the unit vectors $\mathbf{u}_n$ and $\mathbf{u}_t$, respectively.

The center of curvature, O' , always lies on the concave side of the curve. The radius of curvature, ρ , is defined as the perpendicular distance from the curve to the center of curvature at that point.

The position of the particle at any instant is defined by the distance, s , along the curve from a fixed reference point.

VELOCITY IN THE n-t COORDINATE SYSTEM



The **velocity vector** is always tangent to the path of motion (t-direction).

The **magnitude** is determined by taking the **time derivative** of the **path function**, $s(t)$.

$$\mathbf{v} = v \mathbf{u}_t \quad \text{where} \quad v = \dot{s} = ds/dt$$

Here v defines the **magnitude** of the velocity (speed) and $\mathbf{u}_t$ defines the **direction** of the velocity vector.

ACCELERATION IN THE n-t COORDINATE SYSTEM

Acceleration is the time rate of change of velocity:

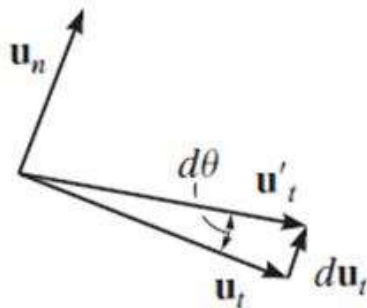
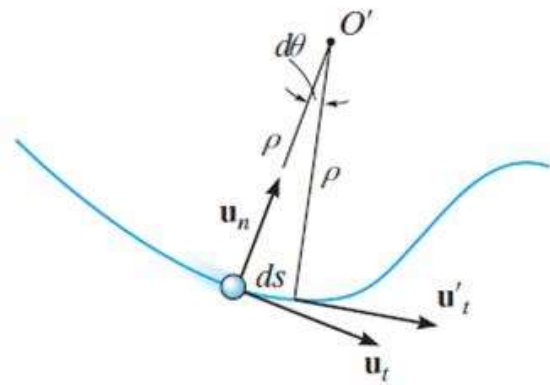
$$\mathbf{a} = d\mathbf{v}/dt = d(v\mathbf{u}_t)/dt = \dot{v}\mathbf{u}_t + v\dot{\mathbf{u}}_t$$

Here $\dot{v}$ represents the change in the magnitude of velocity and $\dot{\mathbf{u}}_t$ represents the rate of change in the direction of $\mathbf{u}_t$.

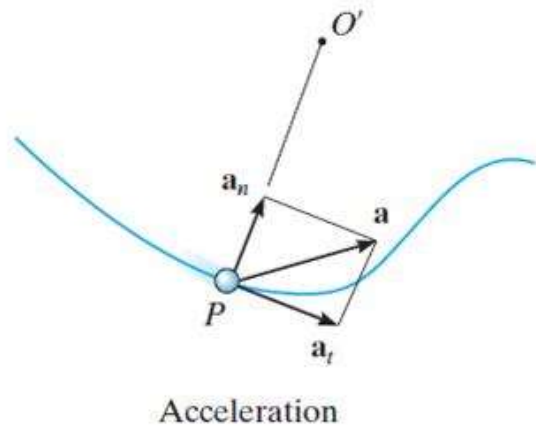
$$\dot{\mathbf{u}}_t = \dot{\theta} \mathbf{u}_n = (\dot{s}/\rho) \mathbf{u}_n = (v/\rho) \mathbf{u}_n$$

After mathematical manipulation, the acceleration vector can be expressed as:

$$\mathbf{a} = \dot{v} \mathbf{u}_t + (v^2/\rho) \mathbf{u}_n = a_t \mathbf{u}_t + a_n \mathbf{u}_n.$$



ACCELERATION IN THE n - t COORDINATE SYSTEM (continued)



So, there are **two** components to the acceleration vector:

$$\mathbf{a} = a_t \mathbf{u}_t + a_n \mathbf{u}_n$$

- The **tangential component** is tangent to the curve and in the direction of increasing or decreasing velocity.

$$a_t = v \dot{\quad} \text{ or } a_t ds = v dv$$

- The **normal** or **centripetal component** is always directed toward the center of curvature of the curve. $a_n = v^2/\rho$

- The **magnitude** of the acceleration vector is

$$a = \sqrt{(a_n)^2 + (a_t)^2}$$

SPECIAL CASES OF MOTION

There are some special cases of motion to consider.

- 1) The particle moves along a **straight line**.

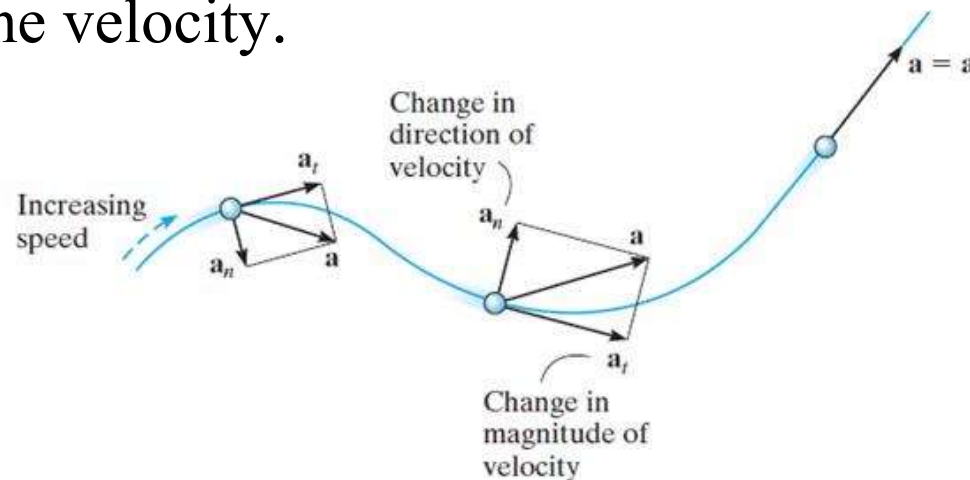
$$\rho \rightarrow \infty \Rightarrow a_n = v^2/\rho = 0 \Rightarrow a = a_t = \dot{v}$$

The **tangential component** represents the **time rate of change** in the **magnitude** of the **velocity**.

- 2) The particle moves along a curve at **constant speed**.

$$a_t = \dot{v} = 0 \Rightarrow a = a_n = v^2/\rho$$

The **normal component** represents the **time rate of change** in the **direction** of the velocity.



SPECIAL CASES OF MOTION (continued)

- 3) The tangential component of acceleration is **constant**,

$$a_t = (a_t)_c.$$

In this case, $s = s_o + v_o t + (1/2) (a_t)_c t^2$

$$v = v_o + (a_t)_c t$$

$$v^2 = (v_o)^2 + 2 (a_t)_c (s - s_o)$$

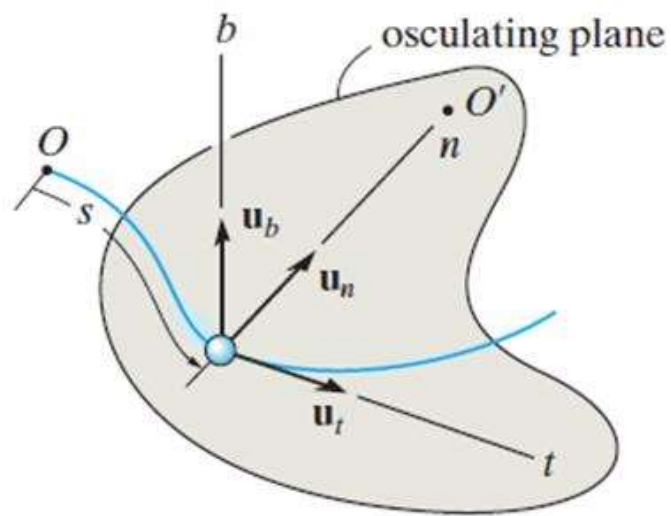
As before, s_o and v_o are the initial position and velocity of the particle at $t = 0$. How are these equations related to projectile motion equations? Why?

- 4) The particle moves along a path expressed as $y = f(x)$.

The **radius of curvature, ρ** , at any point on the path can be calculated from

$$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$$

THREE-DIMENSIONAL MOTION



If a particle moves along a **space curve**, the **n-t** axes are defined as before. At any point, the **t-axis** is **tangent** to the **path** and the **n-axis** points **toward** the **center of curvature**. The plane containing the **n-t** axes is called the **osculating plane**.

A third axis can be defined, called the binomial axis, **b**. The binomial unit vector, $\mathbf{u}_b$, is directed **perpendicular** to the osculating plane, and its **sense** is defined by the **cross product**

$$\mathbf{u}_b = \mathbf{u}_t \times \mathbf{u}_n.$$

There is no motion, thus no velocity or acceleration, in the binomial direction.

PROCEDURE FOR ANALYSIS

Coordinate System.

- Provided the *path* of the particle is *known*, we can establish a set of n and t coordinates having a *fixed origin*, which is coincident with the particle at the instant considered.
- The positive tangent axis acts in the direction of motion and the positive normal axis is directed toward the path's center of curvature.

Velocity.

- The particle's *velocity* is always tangent to the path.
- The magnitude of velocity is found from the time derivative of the path function.

$$v = \dot{s}$$

Tangential Acceleration.

- The tangential component of acceleration is the result of the time rate of change in the *magnitude* of velocity. This component acts in the positive s direction if the particle's speed is increasing or in the opposite direction if the speed is decreasing.
- The relations between a_t , v , t , and s are the same as for rectilinear motion, namely,

$$a_t = \dot{v} \quad a_t ds = v dv$$

PROCEDURE FOR ANALYSIS

- If a_t is constant, $a_t = (a_t)_c$, the above equations, when integrated, yield

$$s = s_0 + v_0 t + \frac{1}{2}(a_t)_c t^2$$

$$v = v_0 + (a_t)_c t$$

$$v^2 = v_0^2 + 2(a_t)_c(s - s_0)$$

Normal Acceleration.

- The normal component of acceleration is the result of the time rate of change in the *direction* of the velocity. This component is *always* directed toward the center of curvature of the path, i.e., along the positive n axis.
- The magnitude of this component is determined from

$$a_n = \frac{v^2}{\rho}$$

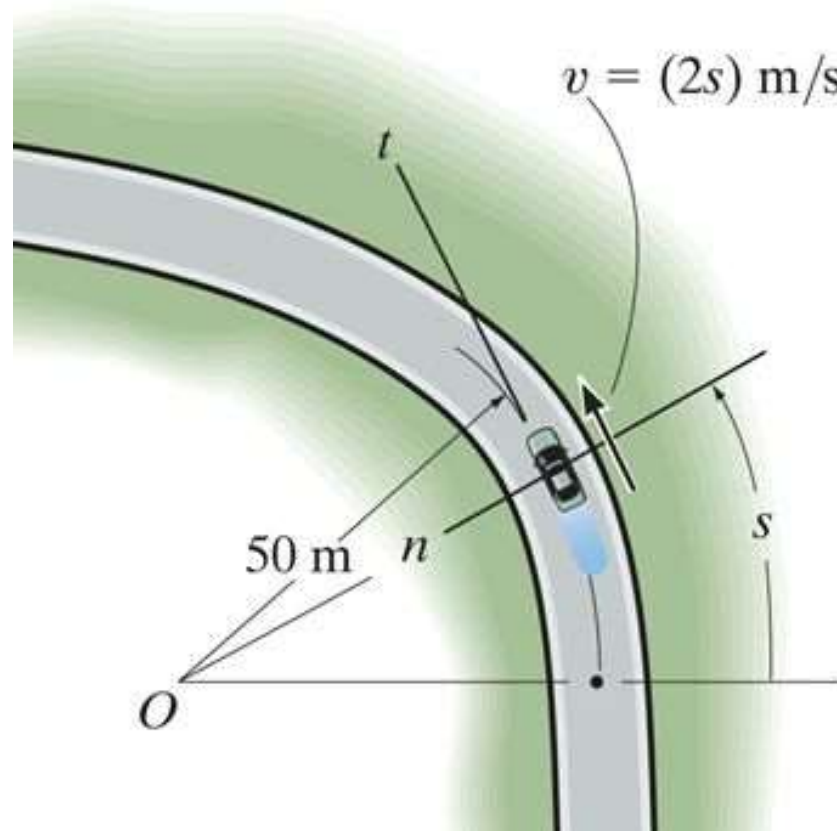
- If the path is expressed as $y = f(x)$, the radius of curvature ρ at any point on the path is determined from the equation

$$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$$

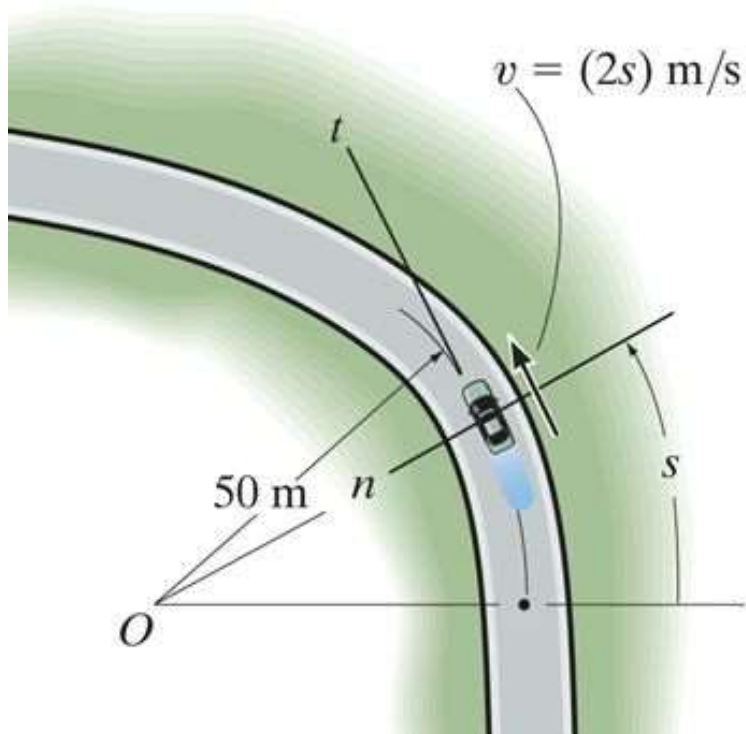
The derivation of this result is given in any standard calculus text.

EXAMPLE I

The car is traveling along the road with a speed of $v = (2s)$ m/s, where s is in meters. Determine the magnitude of its acceleration when $s = 10$ m. **(Problem No. F12-28)**



EXAMPLE I (continued)



Given: A car travels along the road with a speed of $v = (2s)$ m/s, where s is in meters. $\rho = 50$ m

Find: The magnitudes of the car's acceleration at $s = 10$ m.

Plan:

- 1) Calculate the velocity when $s = 10$ m using $v(s)$.
- 2) Calculate the tangential and normal components of acceleration and then the magnitude of the acceleration vector.

EXAMPLE I (continued)

Solution:

- 1) The velocity vector is $\mathbf{v} = v \mathbf{u}_t$, where the magnitude is given by $v = (2s) \text{ m/s}$.

$$\text{When } s = 10 \text{ m: } v = 20 \text{ m/s}$$

- 2) The acceleration vector is $\mathbf{a} = a_t \mathbf{u}_t + a_n \mathbf{u}_n = \dot{v} \mathbf{u}_t + (v^2/\rho) \mathbf{u}_n$

Tangential component:

$$\text{Since } a_t = \dot{v} = dv/dt = (dv/ds) (ds/dt) = v (dv/ds)$$

$$\text{where } v = 2s \Rightarrow a_t = d(2s)/ds (v) = 2 v$$

$$\text{At } s = 10 \text{ m: } a_t = 40 \text{ m/s}^2$$

$$\text{Normal component: } a_n = v^2/\rho$$

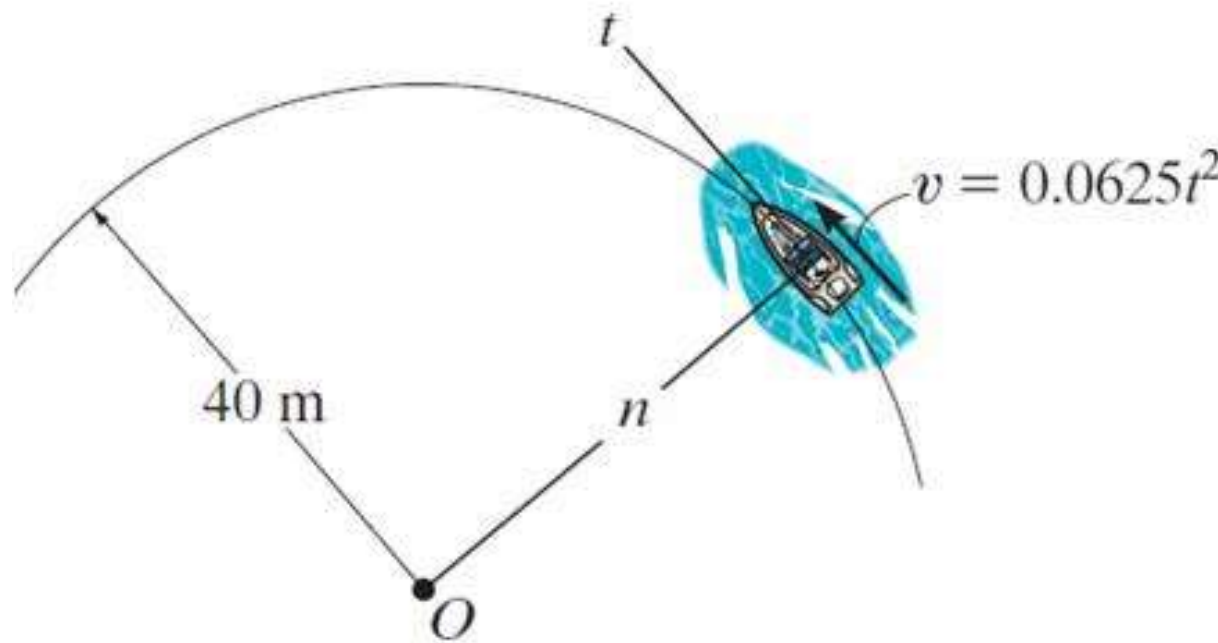
$$\text{When } s = 10 \text{ m: } a_n = (20)^2 / (50) = 8 \text{ m/s}^2$$

The magnitude of the acceleration is

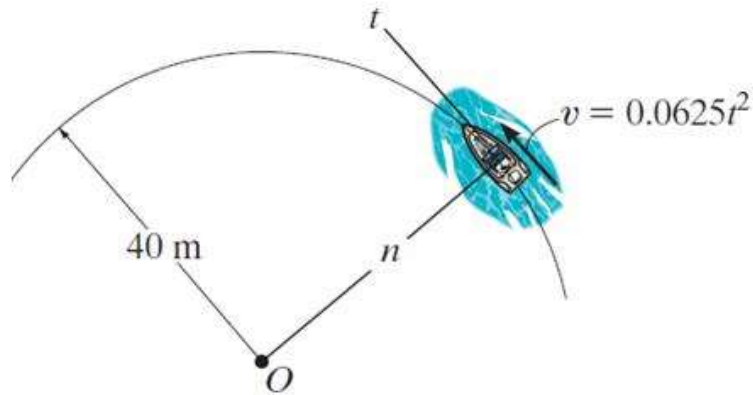
$$a = \sqrt{(a_n)^2 + (a_t)^2} = \sqrt{40^2 + 8^2} = 40.8 \text{ m/s}^2$$

EXAMPLE II

The boat is traveling along the circular path with a speed of $v = (0.0625t^2)$ m/s, where t is in seconds. Determine the magnitude of its velocity and acceleration when $t = 10$ s.
(Problem No. F12-27)



EXAMPLE II



Given: A boat travels around a circular path, $\rho = 40$ m, at a speed that increases with time, $v = (0.0625 t^2)$ m/s.

Find: The magnitudes of the boat's velocity and acceleration at the instant $t = 10$ s.

Plan:

The boat starts from rest ($v = 0$ when $t = 0$).

- 1) Calculate the velocity at $t = 10$ s using $v(t)$.
- 2) Calculate the tangential and normal components of acceleration and then the magnitude of the acceleration vector.

EXAMPLE II (continued)

Solution:

- 1) The velocity vector is $\mathbf{v} = v \mathbf{u}_t$, where the magnitude is given by $v = (0.0625t^2) \text{ m/s}$. At $t = 10\text{s}$:

$$v = 0.0625 t^2 = 0.0625 (10)^2 = \underline{6.25 \text{ m/s}}$$

- 2) The acceleration vector is $\mathbf{a} = a_t \mathbf{u}_t + a_n \mathbf{u}_n = \dot{v} \mathbf{u}_t + (v^2/\rho) \mathbf{u}_n$.

Tangential component: $a_t = \dot{v} = d(.0625 t^2)/dt = 0.125 t \text{ m/s}^2$

At $t = 10\text{s}$: $a_t = 0.125t = 0.125(10) = \underline{1.25 \text{ m/s}^2}$

Normal component: $a_n = v^2/\rho \text{ m/s}^2$

At $t = 10\text{s}$: $a_n = (6.25)^2 / (40) = \underline{0.9766 \text{ m/s}^2}$

The magnitude of the acceleration is

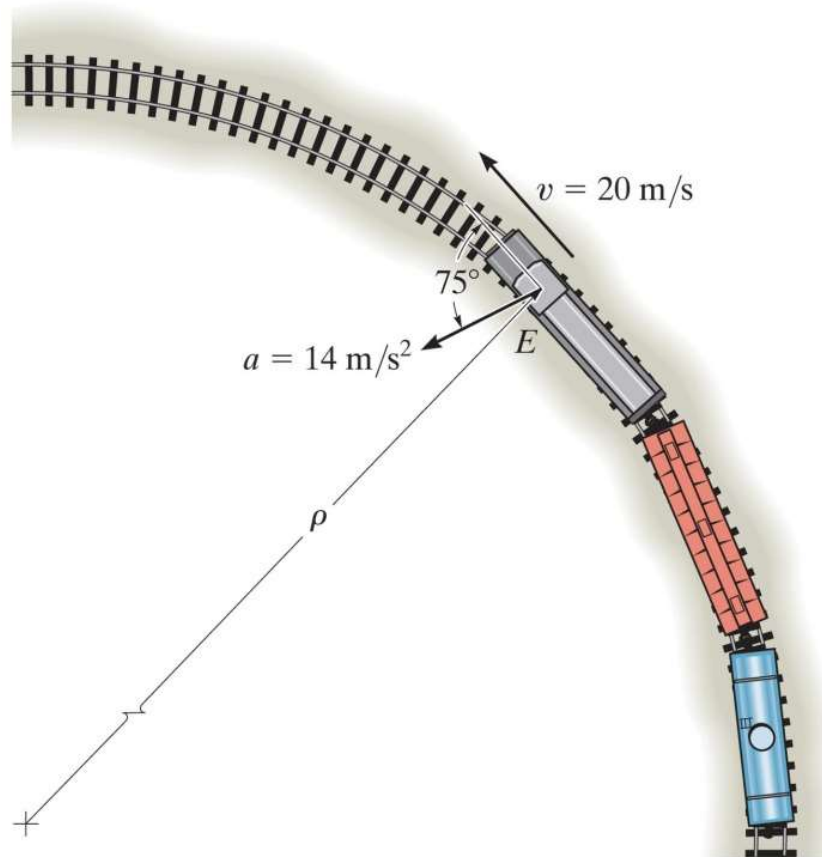
$$a = \sqrt{(a_n)^2 + (a_t)^2} = \sqrt{1.25^2 + 0.9766^2} = \underline{1.59 \text{ m/s}^2}$$

CONCEPT QUIZ

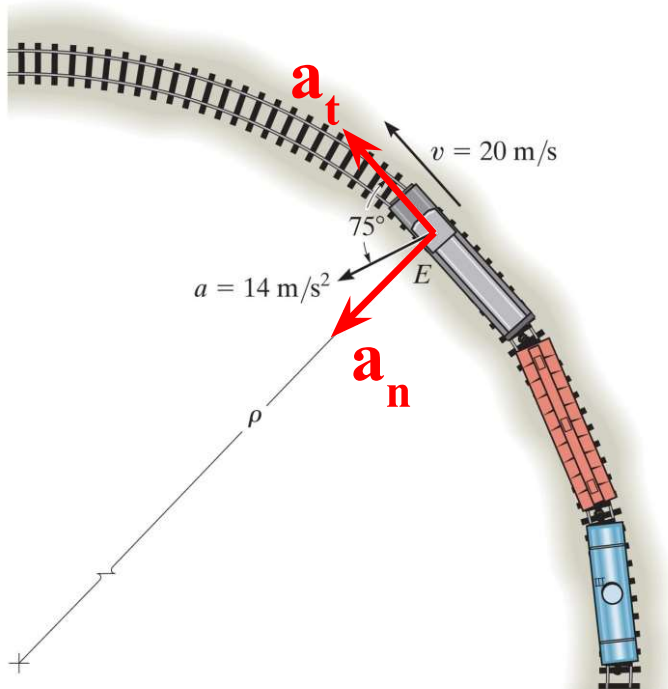
1. A particle traveling in a circular path of radius 300 m has an instantaneous velocity of 30 m/s and its velocity is increasing at a constant rate of 4 m/s^2 . What is the magnitude of its total acceleration at this instant?
A) 3 m/s^2 B) 4 m/s^2
C) 5 m/s^2 D) -5 m/s^2
2. If a particle moving in a circular path of radius 5 m has a velocity function $v = 4t^2 \text{ m/s}$, what is the magnitude of its total acceleration at $t = 1 \text{ s}$?
A) 8 m/s B) 8.6 m/s
C) 3.2 m/s D) 11.2 m/s

GROUP PROBLEM SOLVING I

At a given instant the train engine at E has a speed of 20 m/s and an acceleration of 14 m/s^2 acting in the direction shown. Determine the rate of increase in the train's speed and the radius of curvature r of the path. **(Problem No. 12-127)**



GROUP PROBLEM SOLVING I (continued)



Given: The train engine at E has a speed of 20 m/s and an acceleration of 14 m/s^2 acting in the direction shown.

Find: The rate of increase in the train's speed and the radius of curvature ρ of the path.

Plan:

1. Determine the tangential and normal components of the acceleration.
2. Calculate $\dot{v}$ from the tangential component of the acceleration.
3. Calculate ρ from the normal component of the acceleration.

GROUP PROBLEM SOLVING I (continued)

Solution:

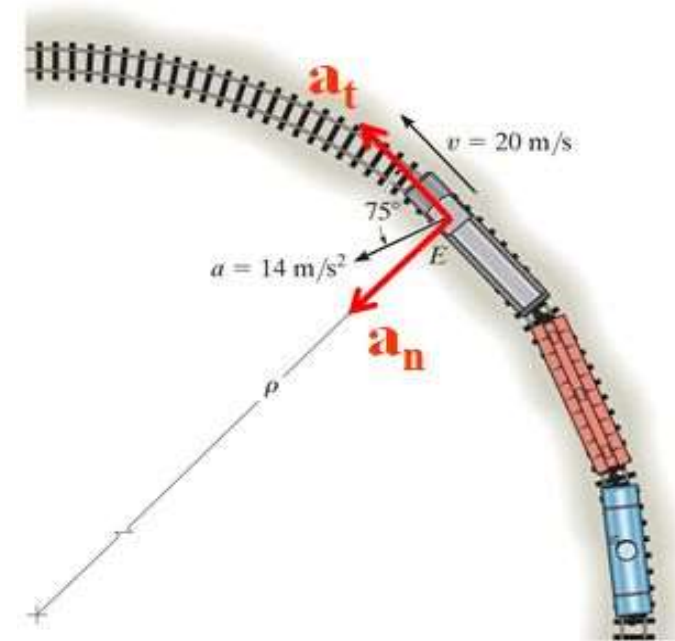
1) Acceleration

Tangential component :

$$a_t = 14 \cos(75) = 3.623 \text{ m/s}^2$$

Normal component :

$$a_n = 14 \sin(75) = 13.52 \text{ m/s}^2$$



2) The **tangential component** of acceleration is the rate of increase of the train's speed, so

$$a_t = \dot{v} = \underline{3.62 \text{ m/s}^2}.$$

3) The **normal component** of acceleration is

$$a_n = v^2/\rho \Rightarrow 13.52 = 20^2 / \rho$$

- $\rho = \underline{29.6 \text{ m}}$

GROUP PROBLEM SOLVING II

Starting from rest, a bicyclist travels around a horizontal circular path, $r = 10$ m, at a speed of $v = (0.09t^2 + 0.1t)$ m/s, where t is in seconds. Determine the magnitudes of his velocity and acceleration when he has traveled $s = 3$ m.
(Problem No. 12-132)

GROUP PROBLEM SOLVING II (continued)

Given: Starting from rest, a bicyclist travels around a horizontal circular path, $\rho = 10$ m, at a speed of $v = (0.09 t^2 + 0.1 t)$ m/s.

Find: The magnitudes of her velocity and acceleration when she has traveled 3 m.

Plan:

The bicyclist starts from rest ($v = 0$ when $t = 0$).

- 1) Integrate $v(t)$ to find the position $s(t)$.
- 2) Calculate the time when $s = 3$ m using $s(t)$.
- 3) Calculate the tangential and normal components of acceleration and then the magnitude of the acceleration vector.

GROUP PROBLEM SOLVING II (continued)

Solution:

- 1) The velocity vector is $v = (0.09 t^2 + 0.1 t)$ m/s, where t is in seconds. Integrate the velocity and find the position $s(t)$.

$$\begin{aligned}\text{Position: } \int v \, dt &= \int (0.09 t^2 + 0.1 t) \, dt \\ s(t) &= 0.03 t^3 + 0.05 t^2\end{aligned}$$

- 2) Calculate the time, t when $s = 3$ m.

$$3 = 0.03 t^3 + 0.05 t^2$$

Solving for t , $t = 4.147$ s

The velocity at $t = 4.147$ s is,

$$v = 0.09 (4.147)^2 + 0.1 (4.147) = \underline{1.96 \text{ m/s}}$$

GROUP PROBLEM SOLVING II (continued)

3) The acceleration vector is $\mathbf{a} = a_t \mathbf{u}_t + a_n \mathbf{u}_n = \dot{v} \mathbf{u}_t + (v^2/\rho) \mathbf{u}_n$.

Tangential component:

$$a_t = \dot{v} = d(0.09 t^2 + 0.1 t) / dt = (0.18 t + 0.1) \text{ m/s}^2$$

$$\text{At } t = 4.147 \text{ s : } a_t = 0.18 (4.147) + 0.1 = \mathbf{0.8465 \text{ m/s}^2}$$

Normal component:

$$a_n = v^2/\rho \text{ m/s}^2$$

$$\text{At } t = 4.147 \text{ s : } a_n = (1.96)^2 / (10) = \mathbf{0.3852 \text{ m/s}^2}$$

The **magnitude** of the acceleration is

$$a = \sqrt{(a_n)^2 + (a_t)^2} = \sqrt{0.8465^2 + 0.3852^2} = \mathbf{0.930 \text{ m/s}^2}$$

ATTENTION QUIZ

1. The magnitude of the normal acceleration is
 - A) proportional to radius of curvature.
 - B) inversely proportional to radius of curvature.
 - C) sometimes negative.
 - D) zero when velocity is constant.
2. The directions of the tangential acceleration and velocity are always
 - A) perpendicular to each other.
 - B) collinear.
 - C) in the same direction.
 - D) in opposite directions.

End of the Lecture
Let The Learning Continue